



B.5 Current and circuits

The particulate nature of matter · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Electric current I is the rate of flow of charge, $I = \Delta Q / \Delta t$, carried by charge carriers (e.g. electrons) drifting through a conductor; resistance $R = V/I$ (Ohm's law: R constant for an ohmic conductor) opposes current flow, with resistivity ρ relating resistance to a conductor's geometry via $R = \rho L/A$. Electrical power dissipated is $P = IV = I^2 R = V^2/R$. In circuits, resistors in series carry the same current and have resistances that add ($R_{\text{total}} = R_1 + R_2 + \dots$), while resistors in parallel have the same voltage across them and combine as $1/R_{\text{total}} = 1/R_1 + 1/R_2 + \dots$; a real cell has internal resistance r , so terminal voltage $V = \mathcal{E} - Ir$, where $\mathcal{E}$ is the electromotive force (emf).

Worked example: follow the reasoning

A cell of emf 6.0 V and internal resistance 0.50 Ω is connected to an external resistor of 5.5 Ω . Find the current in the circuit and the terminal voltage.

Total resistance $= R + r = 5.5 + 0.50 = 6.0 \Omega$.

$I = \mathcal{E} / R_{\text{total}} = 6.0 / 6.0 = 1.0 \text{ A}$.

Terminal voltage $V = \mathcal{E} - Ir = 6.0 - (1.0)(0.50) = 5.5 \text{ V}$.

Answer: $I = 1.0 \text{ A}$, $V = 5.5 \text{ V}$.

Quick check · multiple choice

Two resistors of 4.0 Ω and 12.0 Ω are connected in parallel. What is their combined resistance?

- A 3.0 Ω
- B 8.0 Ω
- C 16.0 Ω
- D 48.0 Ω

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A circuit consists of a battery of emf 12.0 V and internal resistance 1.0 Ω connected to two resistors: a 4.0 Ω resistor in series with a parallel combination of 6.0 Ω and 3.0 Ω resistors. [7]
- (a) Calculate the combined resistance of the parallel section.
 - (b) Calculate the total circuit resistance and the current drawn from the battery.
 - (c) Calculate the power dissipated in the internal resistance of the battery.

Final answer: _____



Answers and explanation

Question 1 [7]

- (a) $1/R_p = 1/6.0 + 1/3.0 = 1/6 + 2/6 = 3/6 = 1/2 \rightarrow R_p = 2.0 \, \Omega$.
(b) Total $R = r + R_{\text{series}} + R_p = 1.0 + 4.0 + 2.0 = 7.0 \, \Omega$. $I = \mathcal{E}/R_{\text{total}} = 12.0/7.0 = 1.71 \, \text{A}$.
(c) $P_{\text{internal}} = I^2 r = (1.71)^2(1.0) = 2.94 \, \text{W}$.

Quick check answer

- A. $1/R = 1/4.0 + 1/12.0 = 3/12 + 1/12 = 4/12 = 1/3 \rightarrow R = 3.0 \, \Omega$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.